Numerical Differentiation

You must complete all the blanks and exercises.

Example Problem

Problem

Estimate the velocity and acceleration from datasets of position of an object.

```
clear all


t = 0:0.2:4;
x = [-5.87 -4.23 -2.55 -0.89 0.67 2.09 3.31 4.31 5.06 5.55 5.78 5.77 5.52 5.08
4.46 3.72 2.88 2.00 1.10 0.23 -0.59];
```

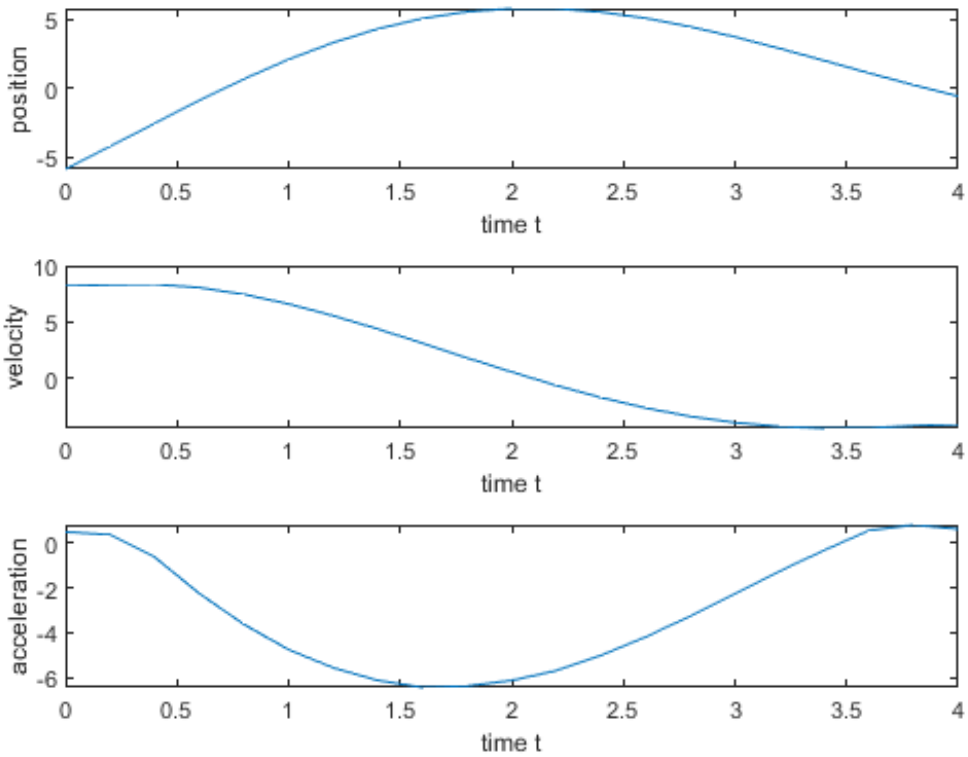
A simple two-point differentiation can be used :

```
vel = derivative(t,x);
acc = derivative(t,vel);


figure
subplot (3,1,1)
    plot(t,x)
    xlabel('time t')
    ylabel('position')


subplot (3,1,2)
    plot(t,vel)
    xlabel('time t')
    ylabel('velocity')


subplot (3,1,3)
    plot(t,acc)
    xlabel('time t')
    ylabel('acceleration')
```



The function `derivative` can be designed simply as

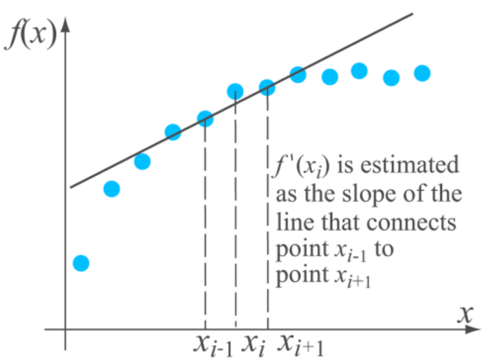
```
function dx = derivative(x,y)
% The derivative: using two-point differentiation of  the central
% finite difference formula, except the first and the last data points.


    n = length(x);
    dx(1)=(y(2)-y(1))/(x(2)-x(1));
    for i=2:n-1
        dydx(i)=(y(i+1)-y(i-1))/(x(i+1)-x(i-1));
    end
    dydx(n)=(y(n)-y(n-1))/(x(n)-x(n-1));
end
```

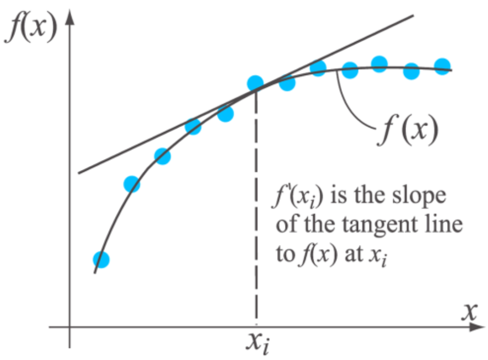
Lecture Objective

This lecture will cover how to calculate a numerical approximation of the derivative from the given discrete datasets.

- **Finite Difference Approximation**
    - Use data points to estimate differentiation.
- Function Approximation
    - From the datasets, apply interpolation or curve-fitting. Then differentiate analytically.



Finite Difference

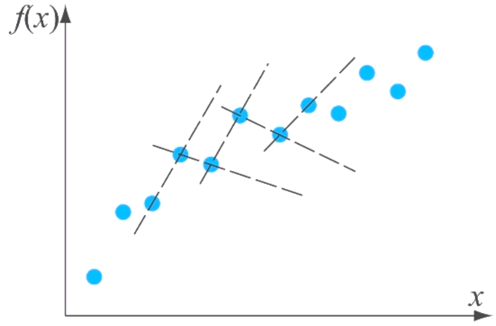


Using Curve fitting

Finite Difference Approximation

We will focus on how to estimate first and the second derivative using the Finite Difference Approximation.

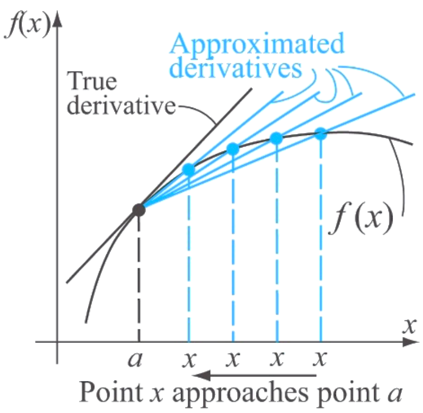
This has some limitation when the datasets contain noises or uncertainty. Even the data should be monotonically increasing, the differentiation results can give negative values. This effect can be reduced if we use a high-order finite differentiation method or the function approximation method.



The derivative f'(x) of a function f(x) at the point x=x0 is defined by:

$$\left.\frac{df(x)}{dx}\right|_{x=x_0} = f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

As x approaches to x0, the accuracy of the derivative increases.



First Derivative

Taylor Series for $f(x+h)$

$$= f(x) + f'(x)h + f''(x)\frac{h^2}{2!} + f^3(x)\cdot\frac{h^3}{3!} + \cdots + f^n(x)\cdot\frac{h^n}{n!} + E$$

Using 2 points

Forward difference:

$$f(x_i+h) = f(x_{i+1}) = f(x_i) + f'(x_i)h + O(h^2)$$

$$f'(x_i)h = f(x_{i+1}) - f(x_i) + O(h^2)$$

$$\boxed{f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{h} + O(h)}$$

Backward:

$$f(x_i-h) = f(x_{i-1}) = f(x_i) + f'(x_i)(-h) + O(h^2)$$

$$f'(x_i)h = f(x_i) - f(x_{i-1}) + O(h^2)$$

$$\boxed{\therefore f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{h} + O(h)}$$

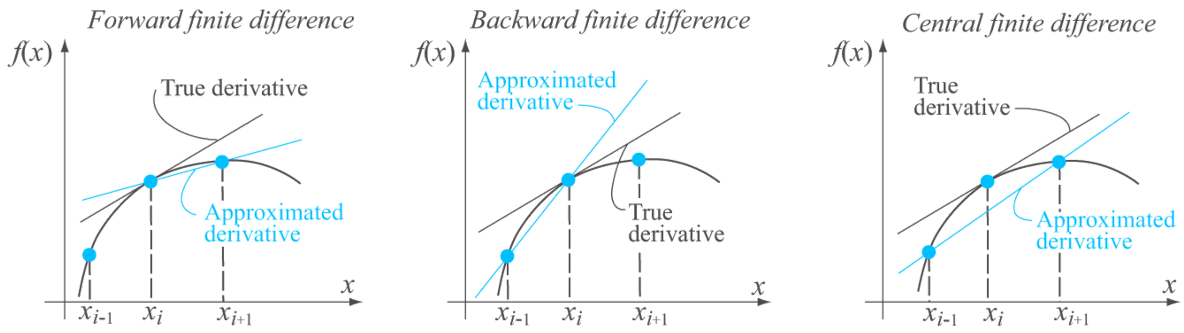
Central Difference:

$$f(x_{i+1}) = f(x_i+h) = \cancel{f(x_i)} + f'(x_i)h + \cancel{f''(x_i)\cdot\frac{h^2}{2!}} + O(h^3)$$

$$- \quad f(x_{i-1}) = f(x_i-h) = \cancel{f(x_i)} - f'(x_i)h + \cancel{f''(x_i)\cdot\frac{h^2}{2!}} - O(h^3)$$

$$f(x_{i+1}) - f(x_{i-1}) = 2\cdot f'(x_i)h + O(h^3)$$

$$\boxed{f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1})}{2h} + O(h^2)}$$



Others

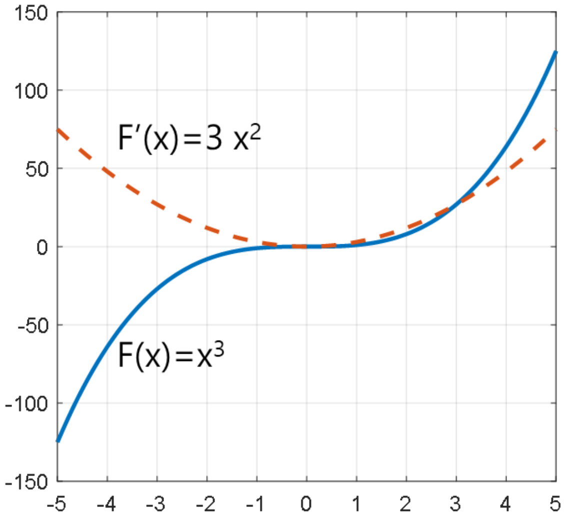
First Derivative		
Method	**Formula**	**Truncation Error**
Two-point forward difference	$f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{h}$	$O(h)$
Three-point forward difference	$f'(x_i) = \frac{-3f(x_i) + 4f(x_{i+1}) - f(x_{i+2})}{2h}$	$O(h^2)$
Two-point backward difference	$f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{h}$	$O(h)$
Three-point backward difference	$f'(x_i) = \frac{f(x_{i-2}) - 4f(x_{i-1}) + 3f(x_i)}{2h}$	$O(h^2)$
Two-point central difference	$f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1})}{2h}$	$O(h^2)$
Four-point central difference	$f'(x_i) = \frac{f(x_{i-2}) - 8f(x_{i-1}) + 8f(x_{i+1}) - f(x_{i+2})}{12h}$	$O(h^4)$

Example: $f(x)=x^3$

Let the data are x =[2, 3, 4] and y=[8, 27, 64].

The true value of $f'(x)= 3x^2$ --> $f'(x)=[12, 27, 48]$ for given x.

2 3 4



At x=3	Forward	Backward	Central
df/dx with h=1	$\frac{f(4)-f(3)}{1} = 64-27 = 37$	$\frac{f(3)-f(2)}{1} = 27-8 = 19$	$\frac{f(4)-f(2)}{2\times1} = \frac{64-8}{2} = 28.$
Error	$\frac{37-27}{27}\times100 = 37.04\%$	$\left\lvert\frac{19-27}{27}\right\rvert\times100 = 29.63\%$	$\frac{28-27}{27}\times100 = 3.70\%$
df/dx with h=0.25	$\frac{f(3.25)-f(3)}{0.25} = \frac{3.25^3-27}{0.25} = 29.3125$	$\frac{f(3)-f(2.75)}{0.25} = \frac{27-2.75^3}{0.25} = 24.8125$	$\frac{f(3.25)-f(2.75)}{2\times0.25} = \frac{3.25^3-2.75^3}{2\times0.25} = 27.0625$
Error	$\frac{29.3125-27}{27}\times100 = 8.56\%$	$\left\lvert\frac{24.8125-27}{27}\right\rvert\times100 = 8.10\%$	$\frac{27.0625-27}{27}\times100 = 0.23\%$

$f'(3)=27$

Derivation of Differentiation

The truncation error can be estimated using Taylor series. *Review Lecture Ch.1*

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1. **Taylor Series for** ***f****(x), where f(x0)is known*

Assumption: $f(x)$ **is differentiable (n+1) times in an interval that contains** x_0

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{(x - x_0)^2}{2!} f''(x_0) + \cdots + \frac{(x - x_0)^n}{n!} f^{(n)}(x_0) + E_{n+1}$$

$$f(x) = \sum_{k=0}^{n} f^{(k)}(x_0) \frac{(x - x_0)^k}{k!} + E_{n+1}$$

$$where,\ E_{n+1} = f^{(n+1)}(\xi) \frac{(x - x_0)^{n+1}}{(n+1)!},\quad \xi\ is\ between\ [x, x_0]$$

2. **Taylor Series for** ***f****(x+h), where f(x) is known.*

$$f(x+h) = f(x) + f'(x)(h) + f''(x)\frac{h^2}{2!} + \cdots + f^{(n)}(x)\frac{h^n}{n!} + E_{n+1}$$

$$= \sum_{k=0}^{n} f^{(k)}(x) \frac{h^k}{k!} + E_{n+1}$$

where,

$$E_{n+1} = f^{(n+1)}(\xi) \frac{h^{n+1}}{(n+1)!} = O(h^{n+1}) \quad , \quad x \le \xi \le x + h,$$

$$|E_{n+1}| \le c|h|^{n+1}$$

1st Derivative

Assumption of equal spacing (h)

$$f(x_i + h) = f(x_i) + f'(x_i)h + \frac{f''(x_i)h^2}{2!} + \cdots$$

2-point forward:

$$f(x_{i+1}) = f(x+h) = f(x_i) + f'(x_i)h + O(h^2)$$

$$f'(x_i)h = f(x_{i+1}) - f(x_i) + O(h^2)$$

$$\boxed{\therefore f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{h} + O(h)}$$

2-point backward:

$$f(x_{i-1}) = f(x_i - h) = f(x_i) + f'(x_i)(-h) + O(h^2)$$

$$f'(x_i)h = f(x_i) - f(x_{i-1}) + O(h^2)$$

$$\boxed{\therefore f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{h} + O(h)}$$

2-point central difference:

$$
-\left|\begin{aligned}
f(x_{i+1}) &= f(x_i+h) = \cancel{f(x_i)} + f'(x_i)h + \cancel{f''(x_i)\frac{h^2}{2!}} + O(h^3) \\
f(x_{i-1}) &= f(x_i-h) = \cancel{f(x_i)} - f'(x_i)h + \cancel{f''(x_i)\frac{h^2}{2!}} - O(h^3)
\end{aligned}\right.
$$

$$f(x_{i+1}) - f(x_{i-1}) = 2f'(x_i)h + O(h^3)$$

$$2hf'(x_i) = f(x_{i+1}) - f(x_{i-1})$$

$$\boxed{\therefore f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1})}{2h}}$$

3-point FW:

$$4f(x_{i+1}) - f(x_{i+2}) = 3f(x_i) + 2f'(x_i)h + O(h^3)$$

$$f'(x_i) = \frac{4f(x_{i+1}) - 3f(x_i) - f(x_{i+2})}{2h} + O(h^2)$$

2^{nd} Derivative

Assumption of equal spacing (h)

3-point central difference

$$+\left\{\begin{aligned} f(x_{i+1}) &= f(x_i) + \cancel{f'(x_i)h} + f''(x_i)\frac{h^2}{2!} + \cancel{f^{(3)}(x_i)\frac{h^3}{3!}} + f^{(4)}(\xi_1)\frac{h^4}{4!} \\ f(x_{i-1}) &= f(x_i) - \cancel{f'(x_i)h} + f''(x_i)\frac{h^2}{2!} - \cancel{f^{(3)}(x_i)\frac{h^3}{3!}} + f^{(4)}(\xi_2)\frac{h^4}{4!} \end{aligned}\right.$$

$$f(x_{i+1}) + f(x_{i-1}) = 2\cdot f(x_i) + f''(x_i)h^2 + o(h^4)$$

$$f''(x_i)h^2 = f(x_{i+1}) - 2f(x_i) + f(x_{i-1}) + o(h^4)$$

$$\boxed{f''(x_i) = \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1})}{h^2} + o(h^2)}$$

Others

Second Derivative		
Method	**Formula**	**Truncation Error**
Three-point forward difference	$f''(x_i) = \dfrac{f(x_i) - 2f(x_{i+1}) + f(x_{i+2})}{h^2}$	$O(h)$
Four-point forward difference	$f''(x_i) = \dfrac{2f(x_i) - 5f(x_{i+1}) + 4f(x_{i+2}) - f(x_{i+3})}{h^2}$	$O(h^2)$
Three-point backward difference	$f''(x_i) = \dfrac{f(x_{i-2}) - 2f(x_{i-1}) + f(x_i)}{h^2}$	$O(h)$
Four-point backward difference	$f''(x_i) = \dfrac{-f(x_{i-3}) + 4f(x_{i-2}) - 5f(x_{i-1}) + 2f(x_i)}{h^2}$	$O(h^2)$
Three-point central difference	$f''(x_i) = \dfrac{f(x_{i-1}) - 2f(x_i) + f(x_{i+1})}{h^2}$	$O(h^2)$
Five-point central difference	$f''(x_i) = \dfrac{-f(x_{i-2}) + 16f(x_{i-1}) - 30f(x_i) + 16f(x_{i+1}) - f(x_{i+2})}{12h^2}$	$O(h^4)$

For data not equally spaced

Use Lagrange polynomial and the differentiate. $(x_i, f(x_i)),\ (x_{i+1}, f(x_{i+1})),\ (x_{i+2}, f(x_{i+2}))$

(Exercise)

- Write down Lagrange polynomial of 2nd order f(x) passing points (x_i, x_{i+1}, x_{i+2})

$$P_n(x) = \sum_{i=0}^{n} y_i L_i \xrightarrow{n=2} P_2(x) = \sum_{i=0}^{2} y_i L_i$$

$$L_i = \prod_{\substack{j=0 \\ j \neq i}}^{n} \frac{x - x_j}{x_i - x_j}$$

$$L_1 = \frac{x - x_i}{x_{i+1} - x_i} \times \frac{x - x_{i+2}}{x_{i+1} - x_{i+2}}$$

$$L_2 = \frac{x - x_i}{x_{i+2} - x_i} \times \frac{x - x_{i+1}}{x_{i+2} - x_{i+1}}$$

$$L_0 = \frac{x - x_1}{x_0 - x_1} \times \frac{x - x_2}{x_0 - x_2} = \frac{(x - x_{i+1})(x - x_{i+2})}{(x_i - x_{i+1})(x_i - x_{i+2})}$$

$$\boxed{\therefore\ P_2(x) = f(x_i) \cdot \frac{(x - x_{i+1})(x - x_{i+2})}{(x_i - x_{i+1})(x_i - x_{i+2})} + f(x_{i+1}) \cdot \frac{(x - x_i)(x - x_{i+2})}{(x_{i+1} - x_i)(x_{i+1} - x_{i+2})} + f(x_{i+2}) \cdot \frac{(x - x_i)(x - x_{i+1})}{(x_{i+2} - x_i)(x_{i+2} - x_{i+1})}}$$

$$\parallel$$

$$f(x)$$

- Then, differentiate the polynomial to get f'(x)

$$f'(x) = \frac{f(x_i)}{(x_i - x_{i+1})(x_i - x_{i+2})}\Big((x - x_{i+2}) + (x - x_{i+1})\Big) + \frac{f(x_{i+1})}{(x_{i+1} - x_i)(x_{i+1} - x_{i+2})}\Big((x - x_{i+2}) + (x - x_i)\Big)$$

$$+ \frac{f(x_{i+2})}{(x_{i+2} - x_i)(x_{i+2} - x_{i+1})}\Big((x - x_{i+1}) + (x - x_i)\Big)$$

$$\boxed{= \frac{f(x_i) \cdot (2x - x_{i+1} - x_{i+2})}{(x_i - x_{i+1})(x_i - x_{i+2})} + \frac{f(x_{i+1}) \cdot (2x - x_i - x_{i+2})}{(x_{i+1} - x_i)(x_{i+1} - x_{i+2})} + \frac{f(x_{i+2})(2x - x_i - x_{i+1})}{(x_{i+2} - x_i)(x_{i+2} - x_{i+1})}}$$

Partial Differentiation

For a function of several independent variables, the partial differentiation of the function with respect to one of the variables represents the rate of the change of the value of the function with respect to this variable.

$$\left.\frac{\delta f(x,y)}{\delta x}\right|_{\substack{x=x_0\\y=y_0}} = \lim_{x\to x_0}\frac{f(x,y_0)-f(x_0,y_0)}{x-x_0}$$

$$\left.\frac{\delta f(x,y)}{\delta y}\right|_{\substack{x=x_0\\y=y_0}} = \lim_{y\to y_0}\frac{f(x_0,y)-f(x_0,y_0)}{y-y_0}$$

First Derivative

Forward difference:

$$\left.\frac{\delta f}{\delta x}\right|_{\substack{x=x_i\\y=y_i}} = \frac{f(x_{i+1},y_i)-f(x_i,y_i)}{x_{i+1}-x_i}$$

$$\left.\frac{\delta f}{\delta y}\right|_{\substack{x=x_i\\y=y_i}} = \frac{f(x_i,y_{i+1})-f(x_i,y_i)}{y_{i+1}-y_i}$$

Backward:

$$\left.\frac{\delta f}{\delta x}\right|_{\substack{x=x_i \\ y=y_i}} = \frac{f(x_i, y_i) - f(x_{i-1}, y_i)}{x_i - x_{i-1}}$$

$$\left.\frac{\delta f}{\delta y}\right|_{\substack{x=x_i \\ y=y_i}} = \frac{f(x_i, y_i) - f(x_i, y_{i-1})}{y_i - y_{i-1}}$$

Central Difference:

$$\left.\frac{\delta f}{\delta x}\right|_{\substack{x=x_i \\ y=y_i}} = \frac{f(x_{i+1}, y_i) - f(x_{i-1}, y_i)}{x_{i+1} - x_{i-1}}$$

$$\left.\frac{\delta f}{\delta y}\right|_{\substack{x=x_i \\ y=y_i}} = \frac{f(x_i, y_{i+1}) - f(x_i, y_{i-1})}{y_{i+1} - y_{i-1}}$$

2nd Derivative

Forward difference:

$$\left.\frac{\delta f^2(x,y)}{\delta x^2}\right|_{\substack{x=x_0\\y=y_0}} = \frac{f(x_{i-1},y_i)-2f(x_i,y_i)+f(x_{i+1},y_i)}{{h_x}^2}$$

$$\left.\frac{\delta f^2(x,y)}{\delta y^2}\right|_{\substack{x=x_0\\y=y_0}} = \frac{f(x_i,y_{i-1})-2f(x_i,y_i)+f(x_i,y_{i+1})}{{h_y}^2}$$

Using 1st order central difference:

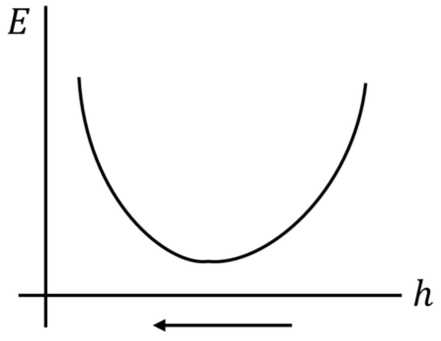
$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)$$

$$= \frac{\partial}{\partial y}\left(\frac{f(x_{i+1},y_i)-f(x_{i-1},y_i)}{2h_x}\right)$$

$$= \frac{\left[f(x_{i+1},y_{i+1})-f(x_{i-1},y_{i+1})\right]-\left[f(x_{i+1},y_{i-1})-f(x_{i-1},y_{i-1})\right]}{2h_x 2h_y}$$

Does the numerical error decreases as h decreases?

The total error E=round_error + truncation_error



$$f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1})}{2h} + O(h^2)$$

$$f'(x_i) = \frac{(f(x_{i+1}) + R_1) - (f(x_{i-1}) + R_2)}{2h} + o(h^2)$$

$$= \frac{f(x_{i+1}) - f(x_{i-1})}{2h} + \underbrace{\frac{R_1 - R_2}{2h}}_{\text{Round error}} + \underbrace{o(h^2)}_{\text{truncation error}}$$

Decreasing h can reduce truncation error, but increase round error.
Therefore, it does not always come to minimize the numerical error.